Calibration and Hedging LAB

Computational Finance 2025/2026

You are provided with a dataset of ICE Brent Crude Oil futures and futures options, covering all quoted bid and ask prices, the daily futures time series, and the SOFR discount curve. The full sample spans **6 January 2020** to **22 December 2022**. You are given the first 15 months (**6 January 2020 – 6 April 2021**) as training data; the remainder (**6 April 2021 – 22 December 2022**) will be released later as an out-of-sample test set (Section 2). Within the training window, you act as a market maker on **6 January 2020**.

1 Task

1. You will price an Asian collar averaging over **February 2021** (Annex A), whose two reference legs are the Apr21 and May21 futures (Annex B). Brent options are American-style but futures-style margined: premiums are marked to market daily, like the underlying future, rather than exchanged at trade date. The payoff described in Annex A references, on each fixing date, the prevailing first-nearby future: Apr21 for most dates, May21 after the roll (Annex B). The reference value is thus a linear combination of two correlated arithmetic averages — a basket option.

2. Extract implied volatilities by inverting the *undiscounted* Black-76 formula (Asay, 1982), i.e. set $r = 0$. State and justify your treatment of the early-exercise premium. Filter out illiquid quotes and calibrate separate smiles for the Apr21 and May21 contracts.

3. Choose a joint dynamics for the two futures. The *baseline* choice is a bivariate, correlated lognormal model (driftless GBMs, Black-76 style); estimate the correlation ρ from historical data (method left to you, but justify it and re-estimate at every recalibration date). As an *optional extension*, you may instead model the dynamics via a jump-extended HJM framework or a spot-driven stochastic volatility model (e.g. Heston, with the futures recovered through cost-of-carry); both typically require pricing by simulation, since closed-form tractability is lost. Calibrate the choosen model on all days the training dataset.

4. Under the baseline lognormal model, price the basket payoff via *one* of the following:

   - *Moment matching*: match the moments of each arithmetic average to an equivalent lognormal variable, then match the moments of their weighted sum, and price via a Black-76-type formula.
   - *Monte Carlo*: simulate your chosen model's dynamics on the fixing dates and evaluate the payoff directly.

   If you adopted an extended model in point 3, Monte Carlo is the natural choice.

   > *Hint: if pricing by simulation, it is worth verifying that your engine converges to the plain-vanilla price on each single underlying, across maturities, before trusting it on the full payoff.*

5. Price the structure in Annex A as of **6 January 2020**: a zero-cost Asian collar with $K_{call} = 78$ fixed; solve for K_{put} such that $\text{Put}(K_{put}) = \text{Call}(78)$ by root-finding on your basket pricer.

6. The market maker sells this collar to the client and is therefore short it, i.e. long $\text{Put}(K_{put})$, short Call(78). Define an automatic hedging strategy, hedging the risk factors you deem relevant (deltas on both legs, vega, ...). You may trade the Apr21/May21 futures and any listed option, accounting for bid-ask spreads, and borrow/lend cash at the SOFR-implied

rate. When hedging with futures or options, you can neglect margin costs, but you must still account for the bid-ask spread.

7. Backtest the strategy from **6 January 2020** to **28 February 2021** (in-sample). At each rebalancing date: recalibrate and reprice; compute the portfolio P&L (structure + hedges + interest + cash); rebalance the hedge.

2 Deliverables

Deliver a report (max 15 pages) and a MATLAB/Python replication package via email to michele.azzone@polimi.it and lucaemanuele.targa@polimi.it. Delivery date **8 December at 1PM**.

After submission, we upload on Webeep the out-of-sample dataset (**6 April 2021 – 22 December 2022**). Over this period, test your strategy — with the *same code* — on the analogous Asian collar averaging over **October 2022** (legs: ICE Brent Dec22 and Jan23 futures, same roll rule as Annex B, same $K_{call} = 78$, K_{put} again solved for zero cost), recalibrating and reporting P&L at every rebalancing date. Deliver a one-page document with: the portfolio value plot, the squared daily P&L plot, and the quantity

$$\frac{1}{T}\sum_{t=1}^{T} \mathrm{P\&L}_t^2,$$

computed on the test set (up until the end of October) — the metric used to rank hedging strategies (lowest wins). Send the replication code. Delivery date for the test set results **11 December** at 11 PM.

3 Evaluation

- **2 points**: quality of the report and of the analysis.
- **Extra points for best hedgers**: 2 points for the top two groups, 1 point for the third and fourth.

Annex A — Training Option Structure Term Sheet

- **Emission Date:** 6 January 2020
- **Notional:** 1,000,000 bbl
- **Reference Value:** ICE Brent Crude futures price (Annex B)
- **Underlying legs:** ICE Brent Apr21 futures (fixings $1, \ldots, n_1$) and ICE Brent May21 futures (fixings $n_1 + 1, \ldots, N$), rolling as per Annex B
- **Structure:** client short Put(K_{put}), long Call($K_{call} = 78$)
- **Strikes:** $K_{call} = 78$ (fixed); K_{put} solved for zero cost at inception
- **Averaging:** arithmetic mean of the Reference Value over the fixing dates in Annex B

Annex B — Training Fixing Rule

The fixing dates are all business days of February 2021. On each such day, the Reference Value is the settlement price of the first-nearby (front-month) ICE Brent Crude futures contract, up to and including the second-to-last fixing day before that contract's expiry; from the next fixing day onward it rolls to the next-nearby contract. The official ICE Brent Crude Futures expiry calendar unambiguously identifies, for every fixing date, whether Apr21 or May21 applies. The same roll logic — applied to October 2022 instead, with Dec22/Jan23 in place of Apr21/May21 — defines the fixing rule for the out-of-sample option structure of Section 2.